[CSE445(5) Summer 2026] Assignment-4: Decision Trees vs. XGBoost

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Abstract

This report details an experimental comparison between Decision Tree and XGBoost models across three machine learning paradigms: Binary Classification, Multi-class Classification, and Multivariate Regression. Hyperparameter tuning was conducted using a strict 70-15-15 sequential train-validation-test split. Evaluation metrics including accuracy, precision, recall, F-score, and mean-squared error were implemented manually without reliance on external metric libraries. The results highlight the performance trade-offs, with XGBoost generally demonstrating superior generalization capabilities.

1 Introduction

Ensemble learning techniques, particularly Gradient Boosted Trees, have become the standard for tabular data tasks. To thoroughly understand their advantages over single tree-based models, this assignment experiments with Decision Tree Classifiers/Regressors and XGBoost across four distinct datasets: BC-BrainCancer for binary classification, UCI Car Evaluation for multi-class classification, and RG-Credit and RG-Wage for regression. Following strict guidelines, the data was partitioned sequentially into 70% training, 15% validation, and 15% testing sets[cite: 1, 2, 3]. Validation sets were exclusively used for hyperparameter tuning to ensure an unbiased evaluation of the test sets.

2 Methodology

2.1 Evaluation Metrics Implementation

A core requirement of this experiment was building a foundational understanding of model evaluation. As such, no standard library metric functions (e.g., from scikit-learn) were used. Instead, a custom suite of metrics was developed using NumPy matrix operations[cite: 1, 2, 3]. For classification, a custom confusion matrix function was designed to track True Positives, False Positives, and False Negatives, from which Accuracy, Precision, Recall, and the F1-score were derived[cite: 1, 2]. For the multi-class dataset, macro-averaging was applied to evaluate performance equitably across minority and majority classes[cite: 2]. For regression tasks, a custom Mean Squared Error (MSE) function was formulated to measure the average squared residuals[cite: 3].

2.2 Trial and Error (Hyperparameter Tuning)

Achieving optimal performance required extensive trial and error. For the Decision Tree models, an exhaustive Grid Search approach was utilized to iterate through combinations of `max_depth`, `min_samples_split`, `min_samples_leaf`, and `criterion`[cite: 1, 2, 3]. Overfitting was a primary concern; for instance, in the Car Evaluation dataset, allowing the tree to grow to a depth of 15 resulted in perfect training accuracy (1.0) but degrading validation accuracy, demonstrating the classic bias-variance trade-off[cite: 2]. Consequently, heavily regularized, shallower trees were often favored.

Given the expansive hyperparameter space of XGBoost, a fixed-seed Random Search was implemented[cite: 1, 2, 3]. We sampled 60 unique combinations across learning rates, estimators, depth, subsampling, and gamma[cite: 1, 2, 3]. Each combination was trained on the training set and evaluated on the validation set, with the highest-scoring combination selected as the final model for the unseen test set[cite: 1, 2, 3].

3 Experiment Results

3.1 Binary Classification: Brain Cancer Dataset

The BrainCancer dataset was used to predict the survival status of patients[cite: 1]. The data was preprocessed by dropping missing values and applying One-Hot Encoding to categorical variables[cite: 1].

Through our tuning process, the best Decision Tree was found at a `max_depth` of 3 using the `gini` criterion, yielding a validation accuracy of 84.62%[cite: 1]. The optimal XGBoost model required a `max_depth` of 5 and a `learning_rate` of 0.01, also achieving 84.62% on the validation set[cite: 1].

However, on the holdout test set, XGBoost demonstrated vastly superior generalization. As shown in Table 1, XGBoost outperformed the Decision Tree across all metrics[cite: 1]. Furthermore, a custom Precision-Recall (PR) curve was generated (Figure 1). The Decision Tree's precision degraded rapidly (Average Precision = 0.444), whereas XGBoost maintained high precision across the recall range (Average Precision = 0.867)[cite: 1].

3.2 Multi-class Classification: Car Evaluation Dataset

The Car Evaluation dataset maps car features to four acceptability classes: unacc, acc, good, and vgood[cite: 2]. Using the macro-averaged custom metrics, we evaluated the models on a highly imbalanced sequential split[cite: 2].

The Decision Tree tuned to a shallow depth of 3 achieved a validation accuracy of 75.68%[cite: 2]. The XGBoost random search selected a `max_depth` of 5 and a `learning_rate` of 0.1, achieving 69.11% on validation[cite: 2]. Interestingly, upon testing, the Decision Tree slightly outperformed XGBoost (Table 2)[cite: 2]. This anomaly is likely due to the sequential splitting method causing a heavily imbalanced and non-representative test set (e.g., zero 'good' class instances in the training set but 46 in the test set)[cite: 2]. The heavily regularized Decision Tree handled this specific distribution shift slightly better.

3.3 Regression: Credit and Wage Datasets

For the regression tasks, we applied a custom MSE function[cite: 3]. In the Wage dataset, columns `logwage` and `region` were proactively dropped to prevent data leakage[cite: 3].

Hyperparameter tuning for regression revealed that XGBoost required relatively small trees (max_depth = 3) to minimize error on both datasets[cite: 3]. The final test results (Table 3) show that XGBoost significantly outperformed the Decision Tree on the Credit dataset (Test MSE of 5181.19 vs. 24064.74) and marginally outperformed it on the Wage dataset (1231.54 vs. 1298.73)[cite: 3].

4 Discussion

The experimental results validate the theoretical strengths of ensemble learning. Through systematic trial and error on the validation sets, we observed that single Decision Trees are highly prone to overfitting, evidenced by our depth-probing tests on the Car dataset[cite: 2]. While enforcing strict regularizations (e.g., max_depth=3) prevents memorization, it severely limits the model's predictive ceiling.

Conversely, XGBoost mitigates this by combining multiple weak learners sequentially. By tuning parameters such as the learning rate and subsampling ratio, XGBoost robustly mapped the feature space, leading to a drastic 2x improvement in Average Precision on the BrainCancer dataset and a roughly 78% reduction in test MSE on the Credit dataset[cite: 1, 3].

Appendix

A. Custom Evaluation Metrics Implementation

The following Python snippet highlights the manual implementation of the macro-averaged F1-score to demonstrate our strict adherence to the constraint against using sklearn's metric library[cite: 2].

```python
def custom_f1(y_actual, y_predicted):
    confusion_matrix = custom_confusion_matrix(y_actual, y_predicted)
    f1_scores = []
    for i in range(len(confusion_matrix)):
        trueP = confusion_matrix[i][i]
        falseP = np.sum(confusion_matrix[:, i]) - trueP
        falseN = np.sum(confusion_matrix[i, :]) - trueP

        precision = 0.0 if trueP + falseP == 0 else trueP / (trueP + falseP)
        recall = 0.0 if trueP + falseN == 0 else trueP / (trueP + falseN)

        if precision + recall == 0:
            f1_scores.append(0.0)
        else:
            f1_scores.append((2 * precision * recall) / (precision + recall))

    return np.mean(f1_scores)
```

Table 1: Binary Classification Test Results (Brain Cancer)

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Decision Tree	0.7143	0.6444	0.6970	0.6500
XGBoost	**0.8571**	**0.8000**	**0.9091**	**0.8250**

Figure 1: Precision-Recall Curve for BrainCancer Dataset (status). XGBoost exhibits significantly higher Area Under the PR Curve (AP=0.867) compared to the Decision Tree (AP=0.444).

Table 2: Multi-class Classification Test Results (Car Evaluation)

Model	**Accuracy**	**Avg Precision**	**Avg Recall**	**Avg F-score**
Decision Tree	**0.6769**	**0.3250**	**0.4931**	**0.3619**
XGBoost	0.6000	0.2469	0.3333	0.2776

Table 3: Regression Test Results (MSE)

Dataset	**Decision Tree (MSE)**	**XGBoost (MSE)**
RG-Credit	24064.74	**5181.19**
RG-Wage	1298.73	**1231.54**